MICRO-PROJECT 13 · COURSE 25DC210

PROBABILITY & STATISTICS

SERVER RESPONSE TIME PREDICTION

Using Regression Models — applying **Probability & Statistics** to model and predict uncertainty in web-service performance.

1,000
OBSERVATIONS

4
WORKLOAD PREDICTORS

R^2 = 0.9505
VARIANCE EXPLAINED

Karunya Institute of Technology and Sciences · School of Computer Science and Technology · Division of Data Science and Cyber Security
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AGENDA

What This Deck Covers

MICRO-PROJECT 13 · 25DC210

01

The Problem

Response time is a random variable, not a constant.

02

Statistical Toolkit

Methods drawn from Syllabus Units IV & V of 25DC210.

03

Dataset & Descriptive Statistics

1,000 time-series observations of server workload.

04

Trend & Correlation

How strongly each workload parameter moves with response time.

05

Regression & Prediction

Multiple Linear Regression, coefficients and error metrics.

06

Validation & Conclusion

Student's t-test, Course Outcome mapping and takeaways.

PROBLEM & RATIONALE

Response Time Is Not a Constant — It Is a Random Variable

Modern web services operate under continuous uncertainty. The response time a client experiences fluctuates heavily with workload and server conditions: **request rate, CPU utilisation, memory pressure and network activity.**

> PROBLEM STATEMENT
>
> How can statistical regression models capture the uncertainty in server response time, predict it from measurable workload parameters, and provide statistical evidence that the model is reliable?

56.7 → 619.2 ms

RESPONSE TIME RANGE

across 1,000 observations — an **~11×** range

FOCUS: statistical reasoning, not black-box AI

The statistical pipeline (Section 1.0 of the project)

01 **Uncertainty**

arrow_downward

02 **Statistical Modelling**

arrow_downward

03 **Correlation**

arrow_downward

04 **Regression**

arrow_downward

05 **Prediction**

arrow_downward

06 **Validation**

SECTION 3.0 · REPORT

Aim & Objectives

1 AIM · 6 OBJECTIVES

AIM

Develop and statistically validate a regression-based model that predicts server response time from relevant workload and system-performance parameters.

01

Collect repeated observations of workload and measured response time

02

Prepare and organise data for statistical analysis

03

Compute descriptive statistics — central tendency and variation

04

Examine trends and measure correlation with workload variables

05

Fit the regression model and predict unseen test observations

06

Evaluate with MAE, MSE, RMSE, R^2 and validate with hypothesis tests

SCOPE — statistical modelling of server response time; software supports the pipeline, but the academic focus stays on statistical reasoning and validation.

Σ SYLLABUS UNITS IV & V

The Statistical Toolkit

5 METHOD FAMILIES

01

Descriptive Statistics

Mean, median, min, max, standard deviation — establish central value & variability.

Data summary

02

Correlation

Pearson correlation; rank correlation when a rank-based view fits better.

Unit IV

03

Regression

$$Y = \beta_0 + \beta_1 X + \varepsilon$$

Multiple regression when several predictors are justified.

Core model

04

Prediction Error

MAE, MSE, RMSE quantify misses; R^2 gives variance explained.

Accuracy

05

Validation

Confidence intervals, Student's t-test, F-test, Chi-square — applied only where assumptions hold.

Unit V

! INSIGHT — Prediction alone proves nothing — reliability must be demonstrated statistically.

SYSTEM PIPELINE

From Raw Observations to Validated Insight

Ten stages — raw telemetry in, statistically validated predictions out. The analytical core (07–10) is highlighted.

01 Server / Web Service

02 Performance Observations

03 Dataset: workload + response

04 Data Preparation

05 Descriptive Stats + Trend

06 Correlation Analysis

07 Regression Model

08 Train (800) / Test (200) split

09 Actual vs Predicted + Error Analysis

10 Validation & Interpretation

1,000 observations

80% train · 20% test

4 explanatory variables

THE DATA

1,000 Time-Series Observations of Server Behaviour

A simulated but statistically-structured dataset: workload variables drive response time with realistic noise (`seed = 42` for reproducibility), plus a slight upward trend engineered over time.

PREDICTOR

Request Rate

50 → 997

req/s

PREDICTOR

CPU Usage %

1.0 → 99.5

%

PREDICTOR

Memory Usage %

MEAN

41.0 ± 12.8

%

PREDICTOR

Network Load

1.0 → 51.8

MB/s

TARGET

Response Time

56.7 → 619.2

ms

RELATIONSHIP ENGINEERED

response ≈ `50 + 2.5·CPU + 1.5·Mem + 3.0·Net` + noise — the model must recover this from data alone

TREND COMPONENT

`+0.05 ms` per time step exercises Unit IV time-series analysis

DESCRIPTIVE STATISTICS · UNIT IV

Response Time: Centre, Spread and Range

MEAN

320.41 ms

STD DEV

118.97 ms

RANGE

56.72 – 619.18 ms

VARIABLE	MEAN	STD DEV	MIN	MAX
Request Rate (req/s)	526.77	276.53	50	997
CPU Usage (%)	42.10	23.69	1.00	99.53
Memory Usage (%)	41.03	12.84	≈ 9.9	≈ 76.9
Network Load (MB/s)	26.32	13.94	1.02	51.83

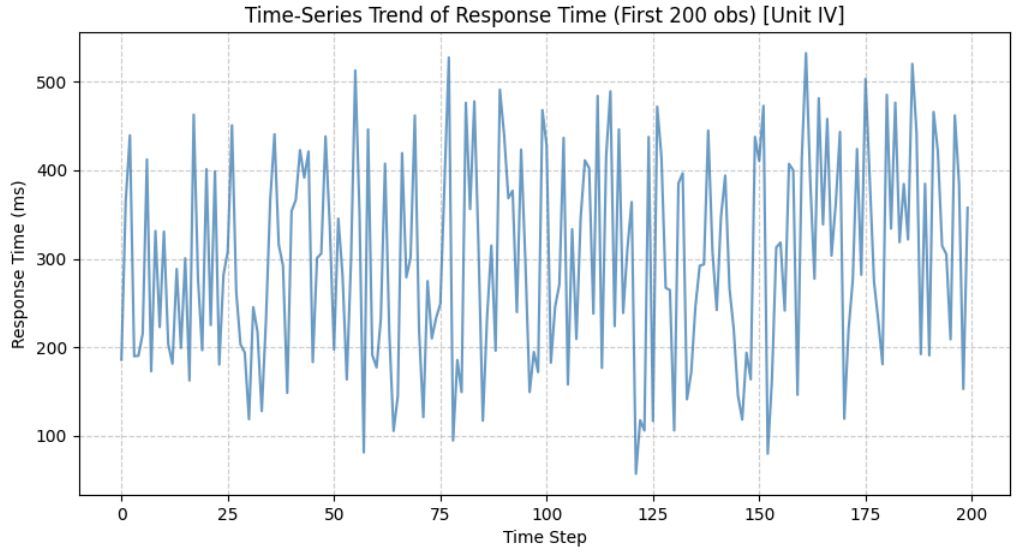
COEFFICIENT OF VARIATION

≈ 37%

Response time is highly dispersed relative to its mean. This is exactly the uncertainty the regression model must explain.

TREND · TIME-SERIES COMPONENT

A Slow Upward Drift Hides in the Noise



Time-series trend of response time — first 200 observations (generated by server_prediction.py)

A secular trend of about `+0.05 ms` per observation was engineered into the series — classic Unit IV trend component

Correlation of `Time_Step` with response time = `0.1249` — weak, but positive and detectable

Detrending matters: ignoring drift would bias the regression residuals

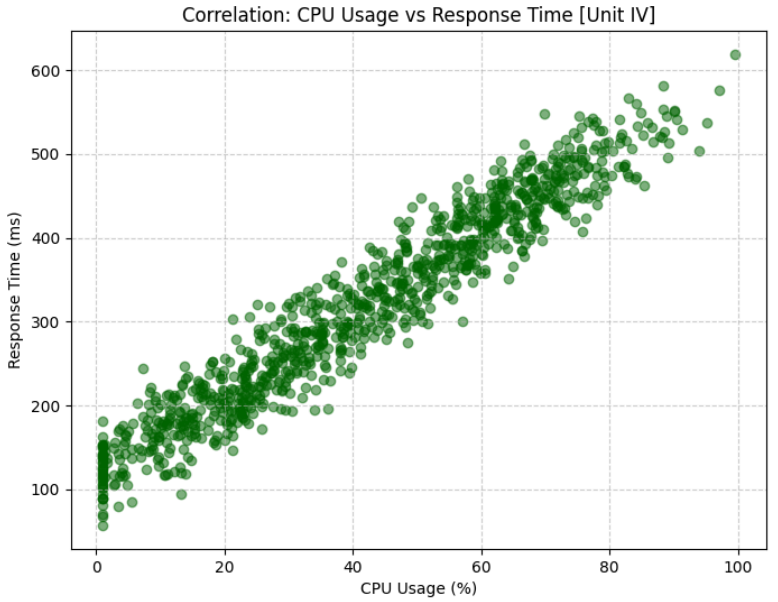
TOOL: time-series decomposition · moving averages (syllabus)

CORRELATION · PEARSON R

Every Workload Parameter Moves With Response Time

CPU Usage	0.9637
Request Rate	0.9423
Network Load	0.9388
Memory Usage	0.9099
Time Step	0.1249

Pearson r with Response_Time_ms, n = 1,000



CPU usage vs response time — near-linear cloud (r = 0.9637)

All four workload predictors are **strongly positively correlated** — good predictive signal, but **collinearity** to keep in mind when reading coefficients.

Σ MODEL · C04 STATISTICAL JUSTIFICATION

Multiple Linear Regression — The Fitted Equation

FITTED MODEL Y = response time · X = workload parameters

$$Response_Time = 73.53 + 0.052 \cdot RequestRate + 2.416 \cdot CPU + 1.447 \cdot Memory + 2.220 \cdot Network$$

β_0 = 73.53 (intercept) · β_1 = 0.052 · β_2 = 2.416 · β_3 = 1.447 · β_4 = 2.220

STRONGEST LEVER

+2.416 ms

per **+1% CPU**

TERM β_2

+2.220 ms

per **+1 MB/s network**

TERM β_4

+1.447 ms

per **+1% memory**

TERM β_3

+0.052 ms

per **+1 req/s**

TERM β_1

Fitted by **ordinary least squares** on **800** training observations; coefficients recover the engineered generative weights **2.5 / 1.5 / 3.0** closely — **the statistics found the true signal.**

C04 — relationship justified by correlation AND regression coefficients

PREDICTION & ERROR EVALUATION · UNIT IV

95% of Response-Time Variance Explained

MAE

20.21 ms

average absolute miss per test observation

RMSE

25.26 ms

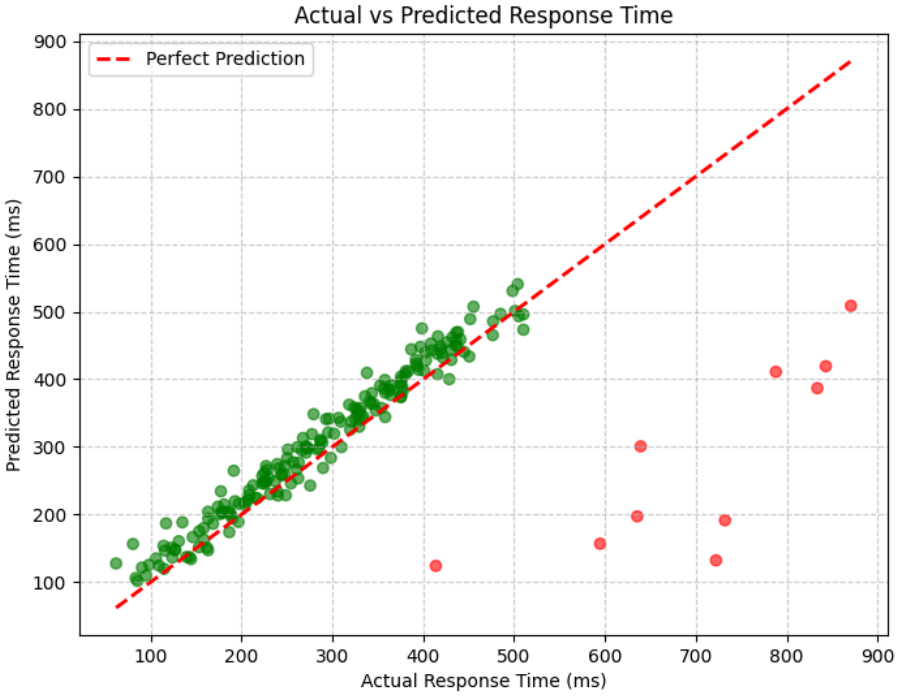
penalises large errors; close to MAE → errors evenly spread, few wild misses

R^2

0.9505

variance explained by the model

200 unseen test observations · 80/20 split



Predicted vs actual response time — points hug the ideal y = x line

RMSE only ~25% above MAE and R^2 near 0.95 — **an honest, well-behaved linear fit** for a noisy system.

HYPOTHESIS TESTING · UNIT V

Is the Model Unbiased? Let the Data Decide

H_0 mean prediction error = 0

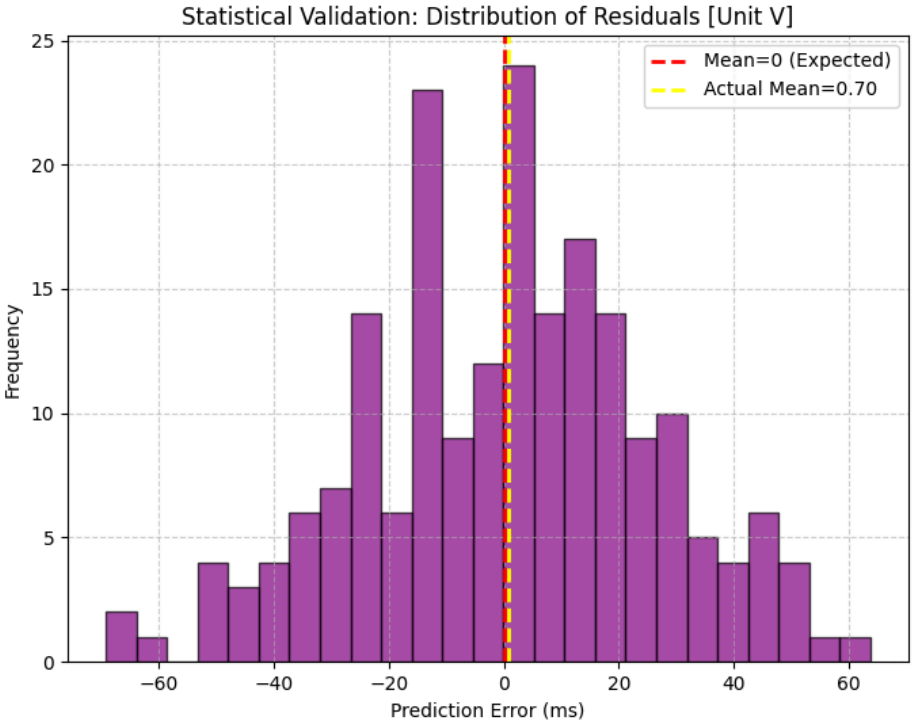
H_1 mean error ≠ 0

One-sample Student's t-test on the **200** test residuals, $\alpha = 0.05$

T-STATISTIC	P-VALUE
0.3931	0.6947

$p = 0.6947 > 0.05 \rightarrow$ **FAIL TO REJECT H_0** — errors are statistically centered on zero: the model is **UNBIASED**

A near-zero t with high p means observed misses look exactly like random noise.



Residual distribution — roughly normal, centered near 0 (red dashed line)

CO5 — validation via RMSE, R^2 AND hypothesis testing

CONCLUSION · CO2–CO5

Statistics Turned Uncertainty Into a Validated Model

WHAT WE SHOWED

- All four workload parameters correlate strongly with response time (r = 0.91 – 0.96)
- The fitted regression explains **95.05%** of response-time variance (R^2 = 0.9505)
- Errors average just **20.21 ms** (MAE) with no systematic bias (t-test p = 0.6947)
- Regression describes association, not causation — model valid within observed workload range

CO2

Modelled uncertainty & variation in response time using workload distributions

02

CO3

Predicted performance under varying, uncertain workload via regression

03

CO4

Justified the relationship with correlation + regression coefficients

04

CO5

Validated with RMSE, R^2 and Student's t-test on residuals

05

APPLICATIONS performance monitoring · capacity planning

FUTURE WORK larger datasets, alternative models, live monitoring dashboard

THANK YOU

Questions & Discussion — happy to walk through the data, the model or the validation.

R^2	MAE	RMSE	P
0.9505	20.21 ms	25.26 ms	0.6947 · unbiased

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